

Abstract

- Food Donation & NGO Connect App** connects food donors with NGOs to manage food donations, requests, inspection, collection, and distribution using **React, FastAPI, and MySQL**.
- It provides a structured and trackable process from **food donation to final distribution**, helping reduce food wastage.

Problem Statement & Objectives

Problem: Manual food donation processes make it difficult to manage and track donations, NGO requests, inspection, and distribution.

Objective 1: Connect food donors with NGOs.

Objective 2: Manage donations, requests, and food inspection.

Objective 3: Track collection and final distribution.

Scope: Donors, NGOs, donation management, inspection, collection, and distribution

Existing vs Proposed System

Feature	Existing System	Proposed System
Donation Management	Manual process	Online donation management
NGO Requests	Phone/messages	Digital request system
Food Inspection	Not systematically recorded	Fresh/Spoiled inspection
Tracking	Difficult to track	Status-based tracking
Limitation / Gap	No centralized workflow	Complete tracking from donation to distribution

Feasibility Study

- Technical:** Uses React, FastAPI, MySQL, VS Code, and Postman.
- Economic:** Uses open-source tools with minimal development cost.
- Operational:** Simple donor/NGO dashboards and easy maintenance.
- Schedule:** Can be completed within the semester timeline.

Results & Testing

- Results:** Successfully manages donations, NGO requests, food inspection, collection, and distribution.
- Testing:** CRUD operations, input validation, authentication, and database connectivity were tested.

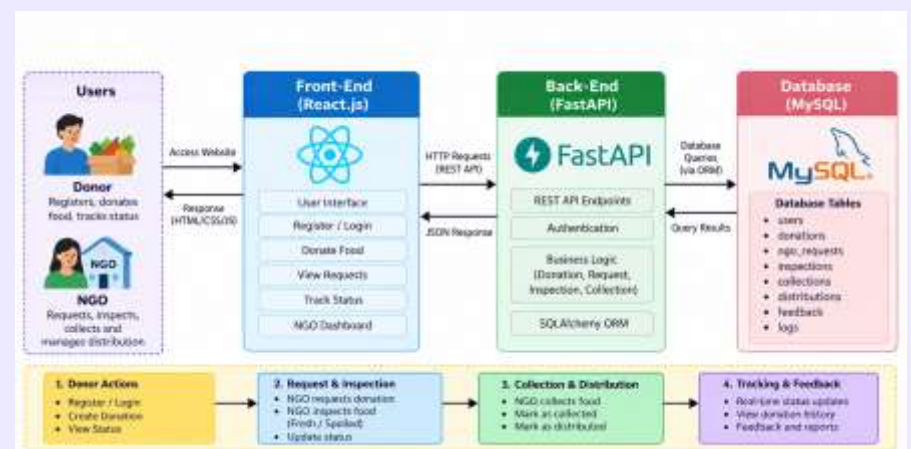
Conclusion

- Achieved:** Developed a web-based platform to manage food donations, NGO requests, inspection, collection, and distribution.
- Objectives Met:** Provides a centralized and trackable workflow connecting donors and NGOs.

Tools & Technologies

- Front-End:** React.js, HTML, CSS, JavaScript
- Back-End:** Python, FastAPI, REST APIs
- Database:** MySQL
- Tools:** VS Code, MySQL Workbench, Postman, Git & GitHub, Draw.io

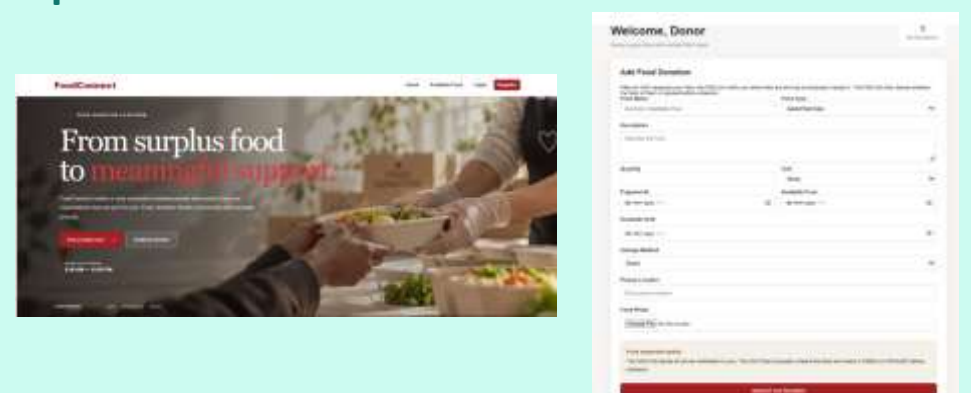
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Future Scope

- Real-time notifications** for donation requests, approvals, and collection updates.
- Location-based matching** to connect nearby donors with NGOs.

